

# CoRoT-2b

R-0105

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_CoRoT-2\_170510 · created 2026-07-16T03:26:29+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                |
|---------------------------------------------------|--------------------------------|
| Mid-Transit Time (Tmid)                           | 2457882.1448 +/- 0.002 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.006 +/- 0.064                |
| Transit depth (Rp/Rs)^2                           | 0.0041 +/- 0.083 [%]           |
| Orbital Inclination (inc)                         | 87.35 +/- 2.06                 |
| Airmass coefficient 1 (a1)                        | 0.8 +/- 0.4                    |
| Airmass coefficient 2 (a2)                        | 0.02 +/- 0.51                  |
| Scatter in the residuals of the lightcurve fit is | 42.8 %                         |
| Best Comparison Star                              | #1 - [204, 411]                |
| Optimal Aperture                                  | 1.58                           |
| Optimal Annulus                                   | 3.0                            |
| Transit Duration (day)                            | 0.019 +/- 0.0047               |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.006 ± 0.064 vs 0.1667 (deviation 2.51σ) |
| Duration fit vs published | 0.019 d vs 0.095 d (deviation 16.17σ)     |
| Mid-time O–C              | -2507.6 min                               |
| Depth SNR                 | 0.1                                       |
| Residual scatter          | 42.8 %                                    |

### Light curve

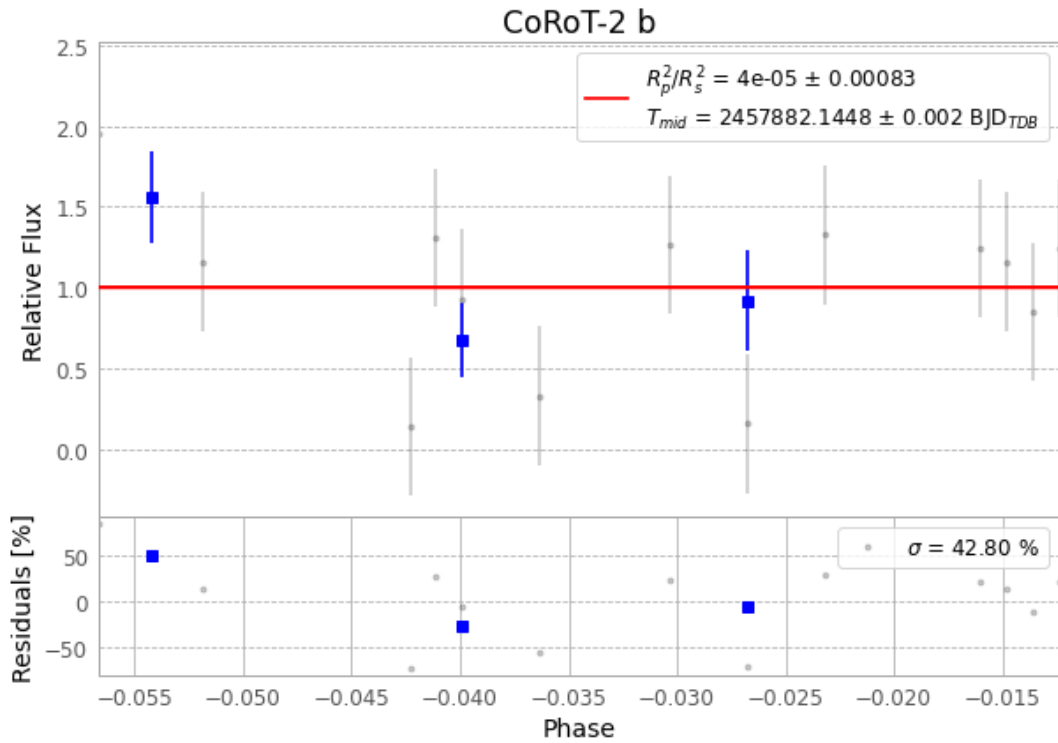


Figure 1 — FinalLightCurve\_CoRoT-2 b\_2017-05-09.png (SHA-256 0a722006c0a2ecf8...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [375, 419], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 5d3ebbbcaa2792de5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

## Data provenance

40 FITS frames (26 MB), CoRoT-2170510065112.FITS → CoRoT-2170510084812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-170510.log (b30f4d998dd3...), FinalLightCurve\_CoRoT-2 b\_2017-05-09.png (0a722006c0a2...), FinalLightCurve\_CoRoT-2 b\_2017-05-09.csv (6590dad4246d...)

## Compute resources

Wall time 110.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 03:26Z.